

HVSr Analysis Report

Synthetic examples · Station SYNA · Recording Site A ambient noise (synthetic) · Generated 2026-09-15 20:59 UTC

SYNTHETIC DATA NOTICE: the analysed recording is a synthetic example, not a field measurement. Results demonstrate the workflow only.

1. Summary

Selected peak frequency f_0	2.470 Hz
Peak amplitude A_0 (H/V)	4.926
Peak selection	automatic
Peak status	clear: Single dominant peak at 2.47 Hz ($A = 4.93$).
Mean curve	geometric — band: $\exp(\text{mean}(\ln \text{HV}) \pm \text{std}(\ln \text{HV}, \text{ddof}=1))$ — log-normal $\pm 1\sigma$ dispersion band
Windows	36 accepted of 39 (3 rejected), 60.0 s, overlap 0.5
SESAME (2004) criteria	reliability PASSED; clear-peak criteria 6/6 (PASSED)
Quality flags	none

2. Station and recording metadata

Project	Synthetic examples
Project description	Demonstration project built from the SYNTHETIC datasets in examples/ (not field measurements).
Station code	SYNA
Station name	Synthetic site A ($f_0 = 2.5$ Hz)
Elevation	152.40 m (500.0 ft)
Recording	Site A ambient noise (synthetic)
Sampling rate (Hz)	100
Start time (UTC)	2025-06-01T08:00:00.000000Z
Duration (s)	1200
Samples per component	120000
Recorded units	counts
Report unit system	Metric (SI)
Synthetic data	yes
Analysed file	/opt/apps/hvsr-studio/data/projects/synthetic-examples-2/runs/01M2K8RA4DDEQDQ77HBNNG727G/processed.npz
Analysed file SHA-256	54fbc5da1b6be8e7903a9aa282d96a47eeba2d65d2ecba2bc7908e0efc2a39a5

Source files

File	Role	Size (bytes)	SHA-256
synthetic_site_A_2p5Hz.txt	all	3935231	049b0ecf1765d00ee50479dc34456d8ab1e8e1d6e78b20a3858363b4831ef6b0

3. Processing settings

Pre-processing steps (applied in order)

#	Operation	Parameters
1	demean	
2	detrend	
3	filter	kind=bandpass; freq_low=0.100; freq_high=40; order=4; zero_phase=yes
4	taper	percent=5

HVSr parameters

Window length / overlap	60 s / 0.500
Per-window detrend / taper	linear / 5 % per side
Smoothing	type=konno_ohmachi; bandwidth=40
Frequency range / points	0.200–30 Hz, 300 log-spaced
Horizontal combination	geometric_mean (after_smoothing)
Mean method	geometric
Peak search	search_min=0.500; search_max=20; min_prominence=0
Vertical floor (masking)	relative=1.000e-06; absolute=1.000e-12
Minimum valid windows per bin	3
Transient screening	enabled=yes; sta_lta=enabled=yes; sta_s=2; lta_s=30; min_ratio=0.200; max_ratio=2.500; amplitude=enabled=no; max_ratio_to_rms=5
Bootstrap	enabled=yes; n_resamples=200
Azimuthal	enabled=yes; step_deg=10

4. Window statistics

Total windows	39
Accepted	36
Rejected	3
Screening enabled	yes
STA/LTA	enabled=yes; sta_s=2; lta_s=30; min_ratio=0.200; max_ratio=2.500
Amplitude criterion	enabled=no; max_ratio_to_rms=5
Masking policy	Per window, bins where the smoothed vertical amplitude $ Z $ is below $\max(1e-12, 1e-06 \times \max Z $ over the analysis band) are excluded (reported as null); bins with fewer than 3 valid windows are invalid in the aggregate curves. Ratios are never clipped.
Invalid frequency bins	0

Rejected windows

#	Start (s)	End (s)	Reason	STA/LTA max	STA/LTA min	Amp. ratio
0	0.0	60.0	sta_lta	3.44	0.77	–
1	30.0	90.0	sta_lta	3.44	0.55	–
38	1140.0	1200.0	sta_lta	1.43	0.00	–

5. Figures

HVSR — geometric mean of 36 accepted windows; band = $\exp(\text{mean}(\ln HV) \pm \text{std}(\ln HV, \text{ddof}=1))$ — log-normal $\pm 1\sigma$ dispersion band

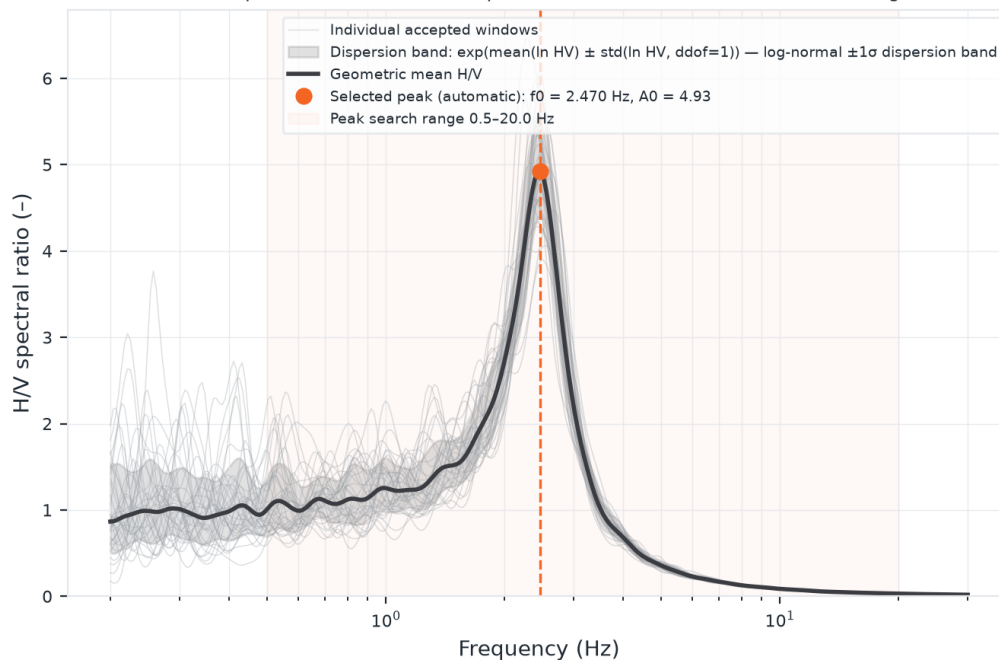


Figure 1. H/V spectral ratio: mean curve, dispersion band, accepted window curves and selected peak.

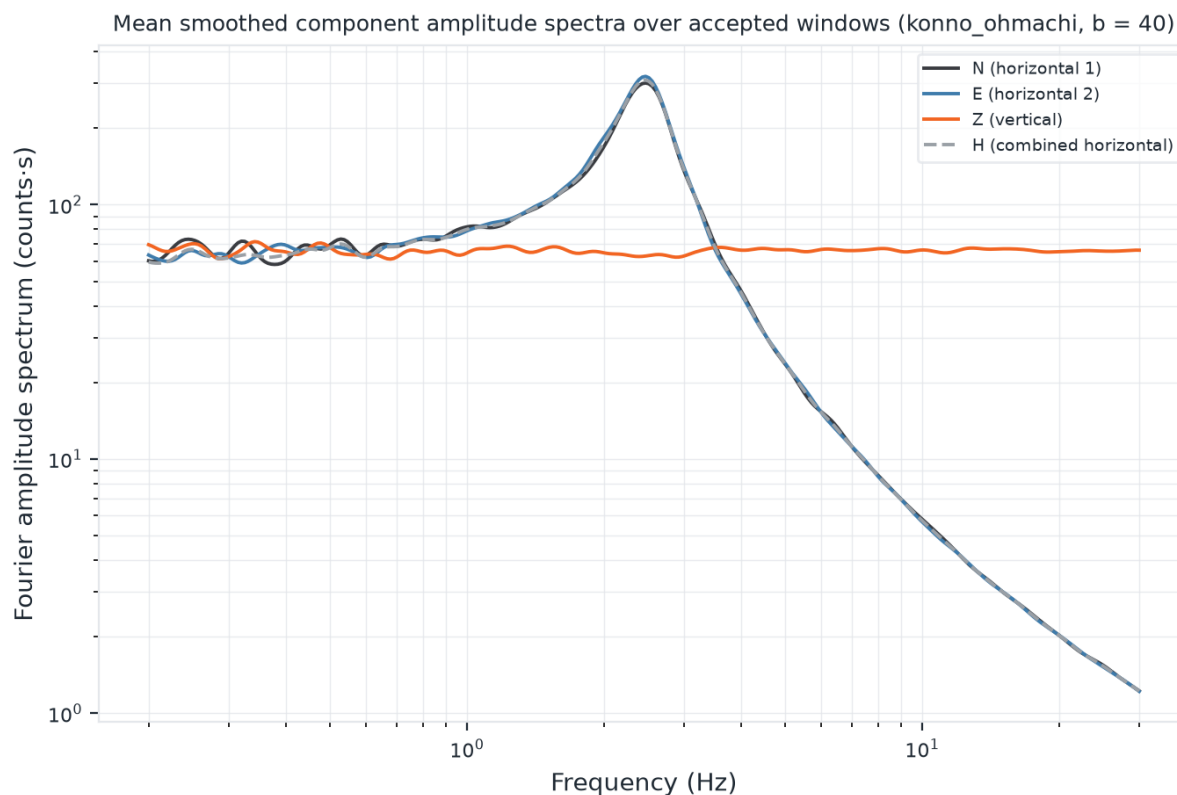


Figure 2. Mean smoothed Fourier amplitude spectra of the three components over accepted windows.

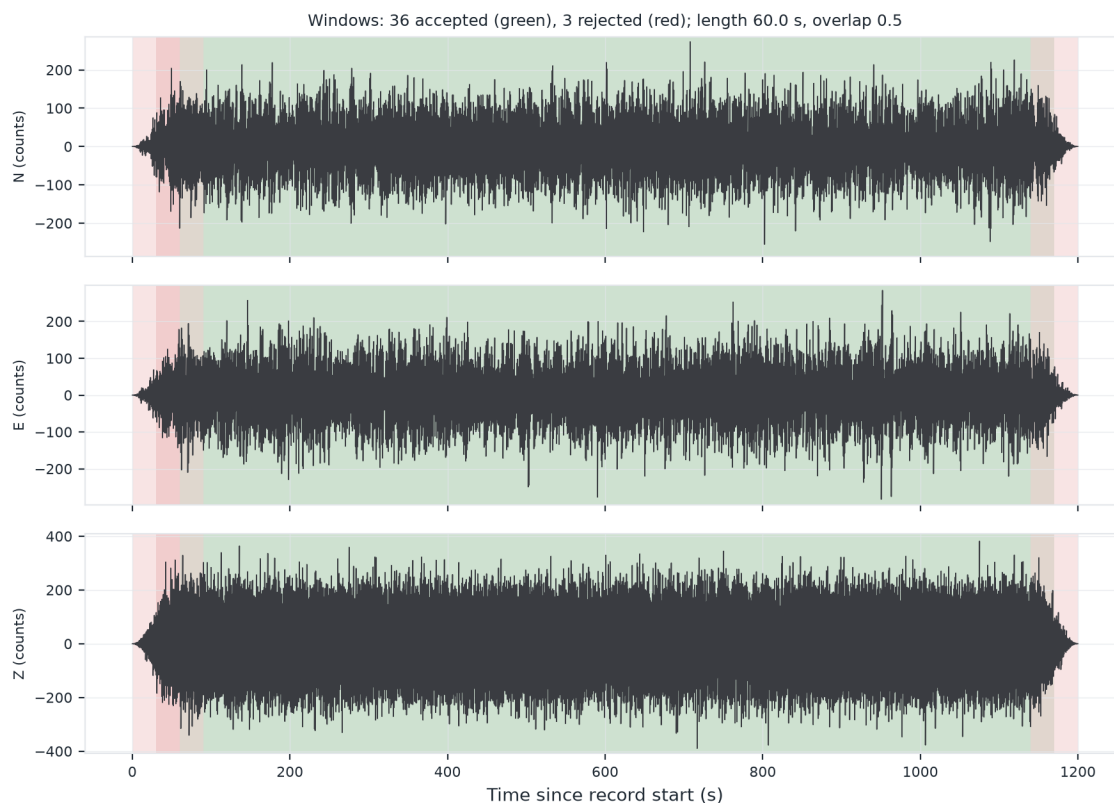


Figure 3. Three-component waveforms with accepted (green) and rejected (red) windows.

Peak frequency of each accepted window curve inside the search range (highest local maximum). std = sample standard deviation (ddof=1) of these frequencies (SESAME σ_f); std_in = standard deviation of $\ln f_0$; p16/p84 = percentiles.

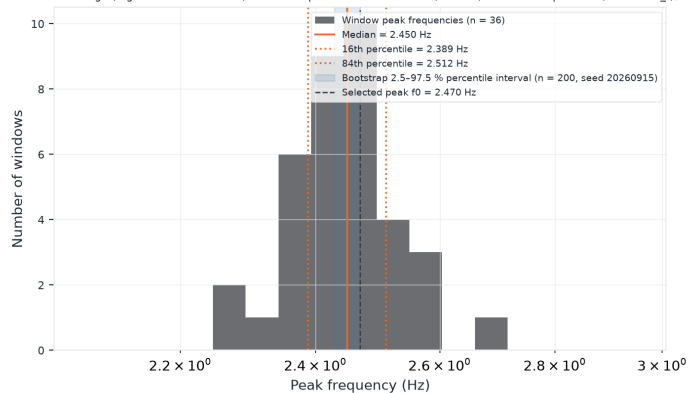


Figure 4. Distribution of window-level peak frequencies with median, 16th/84th percentiles and bootstrap interval.

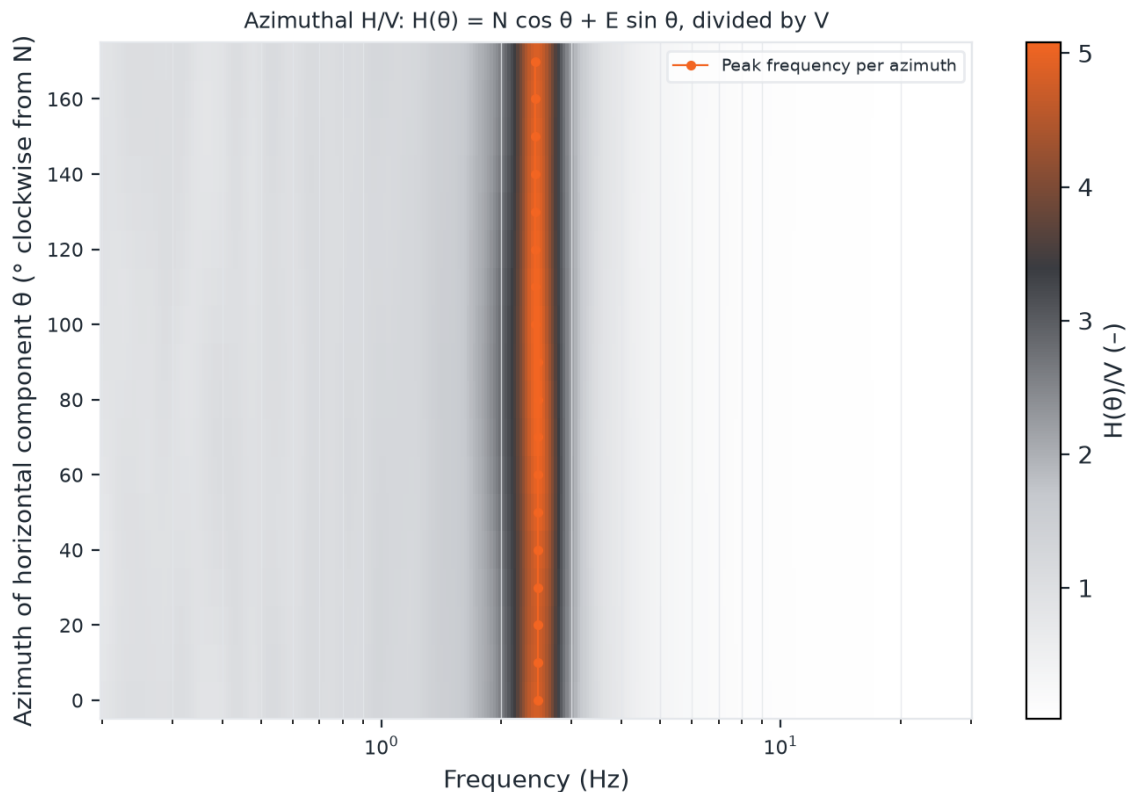


Figure 5. Azimuthal H/V versus frequency and azimuth.

6. Peak and variability

Rank	Frequency (Hz)	Amplitude (H/V)	Prominence
1	2.470	4.926	3.936
2	0.983	1.255	0.033
3	0.817	1.131	0.014
4	0.680	1.129	0.059
5	0.538	1.105	0.084

Search range (Hz)	0.500, 20.000
Selected peak	f0 = 2.470 Hz, A0 = 4.926 (automatic)
Window-level peaks (n)	36
Median f0 (Hz)	2.450
16th / 84th percentile (Hz)	2.389 / 2.512
Std of window f0 (Hz)	0.081
Std of ln f0	0.033
Definition	Peak frequency of each accepted window curve inside the search range (highest local maximum). std = sample standard deviation (ddof=1) of these frequencies (SESAME σ_f); std_ln = standard deviation of ln f0; p16/p84 = percentiles.
Bootstrap	200 resamples, seed 20260915: f0 in [2.429, 2.470] Hz, A0 in [4.753, 5.097] — Bootstrap percentile interval (2.5–97.5 %) of the peak of the mean curve obtained by resampling the accepted windows with replacement 200 times (numpy default_rng, seed 20260915).

7. SESAME (2004) reliability and clear-peak criteria

ID	Criterion	Value	Threshold	Result
R1	$f_0 > 10 / l_w$ — $f_0 = 2.47 \text{ Hz}$ vs $10 / 60 \text{ s} = 0.167 \text{ Hz}$	2.470	0.167	PASS
R2	$n_c(f_0) = l_w \cdot n_w \cdot f_0 > 200$ — $60 \text{ s} \times 36 \text{ windows} \times 2.47 \text{ Hz} = 5335 \text{ cycles}$	5335.416	200.000	PASS
R3	$\sigma_A(f) < 2$ for $0.5f_0 < f < 2f_0$ if $f_0 > 0.5 \text{ Hz}$ ($\sigma_A(f) < 3$ if $f_0 < 0.5 \text{ Hz}$) — $\max \sigma_A$ over $[1.24, 4.94] \text{ Hz} = 1.28$	1.282	2.000	PASS
C1	$\exists f \in [f_0/4, f_0] : A_{H/V}(f) < A_0/2$ — $\min A$ over $[0.618, 2.47] \text{ Hz} = 1.02$ vs $A_0/2 = 2.46$	1.019	2.463	PASS
C2	$\exists f \in [f_0, 4f_0] : A_{H/V}(f) < A_0/2$ — $\min A$ over $[2.47, 9.88] \text{ Hz} = 0.0904$ vs $A_0/2 = 2.46$	0.090	2.463	PASS
C3	$A_0 > 2$ — $A_0 = 4.93$	4.926	2.000	PASS
C4	$f_{\text{peak}}[A_{H/V}(f) \pm \sigma_A(f)] = f_0 \pm 5 \%$ — peak of $\text{mean} \times \sigma_A$ at 2.43 Hz, of mean/σ_A at 2.47 Hz; max deviation 1.7 %	0.017	0.050	PASS
C5	$\sigma_f < \varepsilon(f_0)$ — $\sigma_f = \text{std of 36 window peak frequencies} = 0.081 \text{ Hz}$ vs $\varepsilon(f_0) = 0.05 \cdot f_0 = 0.124 \text{ Hz}$	0.081	0.124	PASS
C6	$\sigma_A(f_0) < \theta(f_0)$ — $\sigma_A(f_0) = \exp(\sigma_{\ln}) = 1.12$ vs $\theta(f_0) = 1.58$ ($\log_{10} \theta = 0.2$)	1.118	1.580	PASS

Reliability: passed (all three criteria required). Clear peak: 6 of 6 criteria fulfilled (passed; at least 5 of 6 required). Criteria are evaluated on the geometric (log) mean curve and its log standard deviation.

Reference: SESAME (2004). Guidelines for the implementation of the H/V spectral ratio technique on ambient vibrations: measurements, processing and interpretation. SESAME European research project, WP12 – Deliverable D23.12, Project No. EVG1-CT-2000-00026, December 2004.

8. Interpretation limitations

- The HVSr amplitude is a spectral ratio of ambient-vibration components. It is NOT a direct measurement of site amplification and must not be used as one.
- The peak frequency f_0 is an estimate of the fundamental resonance frequency of the site under the assumptions of the HVSr technique (SESAME, 2004). No unique subsurface velocity profile can be inferred from an HVSr curve alone.
- Dispersion bands describe the spread of the individual window curves; they are not confidence intervals. The bootstrap interval is a percentile interval of the resampled mean-curve peak.
- Results depend on the processing choices recorded in this file (window length, smoothing, horizontal combination, screening).
- The input is a synthetic example; no geological interpretation applies.

9. Reproducibility

Application version	1.0.0
Engine version	1.0.0
Library versions	engine=1.0.0; python=3.12.14; numpy=2.5.3; scipy=1.18.1; obspy=1.5.1; pandas=2.3.3; matplotlib=3.11.2
Random seed	20260915
Computed at	2026-09-15T19:30:06.258099+00:00
Elapsed (s)	6.401
Analysis directory	/opt/apps/hvsr-studio/data/projects/synthetic-examples-2/analyses/01M2K8RB8R44FW92B3BMAD2453
Result files	result.json, curves.npz, exports/*.csv, exports/summary.json, exports/config.json
Configuration hash (SHA-256/16)	933549fce3b5841c
Report generated	2026-09-15 20:59 UTC

10. Disclaimer

This report was generated automatically by QuakeLogic HVSR Studio from the data, processing settings and window selection described above. The horizontal-to-vertical spectral ratio (HVSR) is an empirical, single-station method: its results depend on the quality of the recording, on ambient-vibration conditions at the time of measurement and on the processing choices recorded in this report. The H/V amplitude is a spectral ratio and must not be interpreted as a direct measurement of site amplification, and an H/V peak frequency does not determine a unique subsurface velocity profile. Interpretation for engineering, geotechnical or regulatory purposes must be performed by a qualified professional, who should verify the data, the processing settings and the reported quality indicators. QuakeLogic Inc. provides this software and the report as is, without warranty of any kind, and accepts no liability for decisions made on the basis of this report. Synthetic data notice: the analysed recording is a synthetic example generated to demonstrate the workflow; it is not a field measurement.

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